



Doc. Ref	TROAX/WMS/042	Date – Rev - 00	28.12.2025	Contract No.	RDJ 188 SCA 042
Project	Construction of East Midfield Cargo Terminal (EMCT) Facility at Zayed International Airport				

PROJECT PLANNING & CPM METHODOLOGY REPORT

Partition Wall Works

Construction of East Midfield Cargo Terminal (EMCT)

Zayed International Airport

Prepared By: Planning Department
Project Duration: 42 Calendar Days
NTP Considered: 01 March 2026
Planned Completion: 12 April 2026



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EXECUTIVE SUMMARY

Project Overview

This report presents the comprehensive Planning, Scheduling, and Execution Strategy for the Partition Wall Installation Works forming part of the Construction of East Midfield Cargo Terminal (EMCT) Facility at Zayed International Airport.

The works are being executed under the requirements of Abu Dhabi Airports (Client) and the project supervision of Hill International (Consultant), in full compliance with statutory, contractual, and safety obligations applicable within the Emirate of Abu Dhabi.

The scope includes the supply, installation, alignment, inspection, and handover of approximately 7,000 square meters of modular partition wall systems (UR-350 and UX-450 configurations), installed in staged vertical segments up to a maximum height of 8,800 mm.

Programme Duration & Strategy

The total planned duration for the partition wall works is 42 calendar days, commencing from 01 March 2026 (Notice to Proceed) and scheduled for completion on 12 April 2026.

The programme has been developed using the Critical Path Method (CPM) and structured within Primavera P6, ensuring a logic-driven, resource-loaded, and compliance-integrated schedule.

The execution strategy is based on:

- Fast-track parallel zone execution
- Overlapping procurement and site works
- Height-wise staged installation methodology
- Embedded quality and safety hold points
- Dedicated installation crews per zone
- Continuous monitoring and weekly schedule updates

The programme adopts an aggressive but structured approach, achievable under controlled and coordinated execution conditions.



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Planning Methodology

The schedule has been developed following industry-standard CPM principles, incorporating:

- Work Breakdown Structure (WBS) aligned with execution zones
- Logical sequencing of activities using FS/SS relationships
- Integration of procurement, installation, inspection, and handover activities
- Resource allocation and load balancing
- Float analysis and critical path identification
- Risk-sensitive activity sequencing

Primavera P6 has been utilized as the primary scheduling platform to:

- Define project calendars
- Assign resources and cost loading (if required)
- Identify critical activities
- Generate lookahead schedules
- Monitor actual vs planned progress
- Facilitate delay analysis if required

Integration of Work Method Statement (WMS)

The Work Method Statement is fully embedded into the schedule logic rather than treated as a standalone procedural document.

The installation has been divided into three controlled height stages:

1. 0-2,200 mm (Ground Level Installation)
2. 2,200-4,400 mm (Movable Scaffold Installation)
3. 4,400-8,800 mm (MEWP / Manlift Installation)



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Each stage includes:

- Permit to Work requirements
- Safety controls and equipment inspections
- QA/QC hold points as per ITP
- Height-specific risk mitigation measures
- Inspection and approval gates prior to progression

This ensures that safety, quality, and execution compliance are structurally linked to the project timeline.

Critical Path & Schedule Sensitivity

Preliminary CPM analysis indicates that the probable critical path flows through:

- Material procurement and delivery
- Final installation zones (notably 6003-RB, 6004-RB, and high-level sections)
- MEWP-based high-level installations
- Final inspection, snag rectification, and MIR approval
- Handover milestone

Activities related to high-level installation and final approvals demonstrate minimal float and therefore require strict monitoring.

The programme is particularly sensitive to:

- Delays in material supply
- Area handover readiness (FFL completion)
- Permit approvals
- Inspection turnaround times

Any disruption in these areas may directly affect the planned completion date.



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Safety & Regulatory Compliance

All works are executed in full compliance with:

- ADOSH Codes of Practice
- Airport operational safety rules
- Approved MSRA and ITP documents
- Permit to Work procedures
- MEWP and scaffolding certification requirements

High-risk activities such as work at height and lifting operations are governed by:

- Certified equipment usage
- Trained and licensed operators
- Exclusion zones
- Fall protection systems
- Daily inspection protocols

The planning structure ensures no high-risk activity proceeds without permit validation and safety clearance.

Quality Assurance Integration

Quality control is not treated as a post-construction activity but is integrated within the schedule through:

- Inspection milestones
- Hold points before vertical progression
- Joint treatment and finishing sequences
- Snag rectification timelines



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- MIR submissions aligned with zone completion

This approach minimizes end-of-project congestion and ensures systematic closeout.

Risk Management Approach

The schedule has been prepared under an optimistic but controlled execution assumption. Risk mitigation strategies embedded within the programme include:

- Parallel crew deployment
- Zone-based sequencing
- Flexible resource reallocation
- Night shift option (if approved)
- Recovery planning through activity crashing

Continuous monitoring through Primavera P6 allows early detection of deviations and proactive corrective action.

Planning & Scheduling Methodology - Primavera P6 Framework

The project schedule has been developed using **Critical Path Method (CPM)** within **Primavera P6**, ensuring a structured, logic-driven, and resource-integrated programme.

Primavera P6 has been utilized as the primary planning platform due to its capability to:

- Develop structured Work Breakdown Structures (WBS)
- Establish logical relationships between activities
- Identify critical path and float
- Allocate and level resources
- Integrate safety and quality milestones



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- Monitor and update progress systematically

The schedule is prepared following internationally recognized CPM principles and aligned with airport operational constraints.

Work Breakdown Structure (WBS) Methodology

The Work Breakdown Structure (WBS) forms the backbone of the Primavera schedule. It divides the project into manageable control sections aligned with execution strategy.

Level 1 - Project Level

EMCT - Partition Wall Works

Level 2 - Major Phases

- Pre-Construction
- Procurement
- Site Execution
- Quality Control
- Testing & Handover

Level 3 - Zone-Based Execution

Site Execution is further broken down into area-wise control zones:

- 6005-RB
- 6006-RB
- 6007-RB
- 6010-RB
- 6011-RB
- 6008-RB
- 6009-RB



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- 6001-RB
- 6002-RB
- 6003-RB
- 6004-RB
- M1-X-6001-RB
- M2-X-6002-RB

This zone-wise WBS allows parallel execution, progress tracking per area, and identification of area-specific delays.

Activity Development Methodology

Within each WBS zone, activities are structured vertically based on the approved Work Method Statement (WMS) installation stages:

Stage 1 - Ground Level Installation (0-2,200 mm)

- Setting out & alignment
- Post fixing
- Panel installation
- QA inspection

Stage 2 - Scaffold Level Installation (2,200-4,400 mm)

- Scaffold erection & inspection
- Work-at-height permit
- Upper framing installation
- Panel installation
- QA inspection

Stage 3 - MEWP Level Installation (4,400-8,800 mm)



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- MEWP mobilization & inspection
- Lifting permit approval
- High-level framing
- Panel installation
- Final QA inspection

Close-Out Activities

- Joint treatment
- Snag rectification
- MIR submission
- Client inspection
- Handover milestone

This structured breakdown ensures full traceability between schedule and method statement.

Logic Development in Primavera

Logical relationships between activities were created using CPM dependency types:

- Finish-to-Start (FS) for sequential operations
- Start-to-Start (SS) for parallel execution
- Inspection hold points inserted as mandatory predecessors

Example Logic Flow:

Ground Installation → Scaffold Installation → MEWP Installation → Inspection → Snag Rectification → MIR → Handover

Procurement activities are linked to installation start to ensure material availability.



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Resources

The assigned resources for the Partition Wall Works at Zayed International Airport have been strategically structured and resource-loaded within the Primavera P6 schedule to support the fast-track 42-day execution programme while maintaining strict compliance with airport operational, safety, and quality requirements under the supervision of Abu Dhabi Airports and Hill International. The resource framework is categorized into management and supervision staff, execution crews, quality and safety personnel, and plant and equipment resources, all integrated within the Critical Path Method (CPM) logic to ensure balanced allocation and controlled productivity. At the management level, the Project Manager retains overall responsibility for programme delivery, stakeholder coordination, and contractual compliance, while the Construction Manager and Site Engineers manage day-to-day site operations, area handovers, permit coordination, and productivity tracking; the Planning Engineer maintains and updates the Primavera schedule, performs critical path analysis, monitors float consumption, and generates progress reports and lookahead programmes. These supervisory roles are generally assigned as level-of-effort resources spanning the project duration to ensure uninterrupted oversight. Execution resources are deployed through multiple parallel crews to meet the aggressive timeline, with specialized teams assigned according to installation height stages defined in the Work Method Statement. Ground-level teams (0-2,200 mm) consist of skilled partition installers and helpers responsible for marking, post fixing, panel alignment, and initial quality checks, operating across multiple zones simultaneously to accelerate progress. Scaffold-level teams (2,200-4,400 mm) include certified scaffold erectors and trained installers authorized for work-at-height activities, ensuring scaffold erection, inspection, green tagging, and safe installation of upper framing components before progressing further. For high-level installations (4,400-8,800 mm), licensed MEWP operators, experienced installers, and banksmen are deployed, recognizing that MEWP availability and safety clearance are critical constraints that directly influence the project's critical path. Dedicated QA/QC engineers and inspectors are embedded within the schedule as mandatory inspection points aligned with the Inspection and Test Plan (ITP), ensuring dimensional accuracy, plumb verification, joint treatment inspection, and material compliance prior to



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progression to subsequent stages, while HSE managers and safety officers oversee MSRA implementation, toolbox talks, permit-to-work validation, scaffold inspection certification, and compliance with airport and ADOSH safety standards. Plant and equipment resources—including movable scaffolding towers, MEWPs, laser leveling instruments, drilling tools, and lifting accessories—are defined within Primavera as controllable assets, with equipment allocation monitored through resource histograms to prevent overallocation and idle time. The scheduling strategy emphasizes resource optimization through zone-wise shifting of crews upon completion of activities, prioritization of critical activities with zero float, and dynamic reallocation during weekly progress updates to mitigate potential delays caused by material supply constraints or area readiness issues such as FFL completion and block wall marking. By integrating manpower, supervision, safety oversight, and equipment planning into a single CPM-based Primavera framework, the project ensures transparency, accountability, and proactive schedule control, thereby safeguarding the targeted completion milestone while maintaining the highest standards of safety, quality, and operational coordination within a live airport environment.

Conclusion

The 42-day programme for partition wall works has been developed using internationally recognized CPM principles and Primavera P6 scheduling methodology, fully aligned with the approved Work Method Statement and regulatory framework governing airport construction projects in Abu Dhabi.

The schedule is technically achievable provided:

- Timely procurement
- Prompt area handover
- Strict safety compliance
- Coordinated quality inspections
- Effective resource management



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The integrated planning approach ensures that execution, safety, quality, and compliance operate as a unified system rather than independent processes, thereby supporting successful and timely project delivery.